

Name:

Key

4. a) Use row reduction to put the following matrix in reduced row echelon form.

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 12 \\ 3 & 9 & 12 \end{pmatrix} \xrightarrow{\frac{1}{3}R_2} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 4 \\ 3 & 9 & 12 \end{pmatrix} \xrightarrow{\frac{1}{3}R_3} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 4 \\ 1 & 3 & 4 \end{pmatrix} \xrightarrow{R_3-R_1} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 4 \\ 0 & 1 & 4 \end{pmatrix}$$

$$\xrightarrow{R_3-R_2} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 4 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1-2R_2} \begin{pmatrix} 1 & 0 & -8 \\ 0 & 1 & 4 \\ 0 & 0 & 0 \end{pmatrix}$$

b) Write the following linear system over $\mathbb{R}$ as an augmented matrix. Then find the general solution to the system and write it in vector form.

$$\begin{aligned} x + 2y - 3z &= -13 \\ x &- 4z = -30 \\ x + 4y - 2z &= 4 \end{aligned}$$

$$\left(\begin{array}{ccc|c} 1 & 2 & -3 & -13 \\ 1 & 0 & -4 & -30 \\ 1 & 4 & -2 & 4 \end{array} \right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{ccc|c} 1 & 0 & -4 & -30 \\ 1 & 2 & -3 & -13 \\ 1 & 4 & -2 & 4 \end{array} \right)$$

$$\xrightarrow{R_2-R_1} \left(\begin{array}{ccc|c} 1 & 0 & -4 & -30 \\ 0 & 2 & 1 & 17 \\ 1 & 4 & -2 & 4 \end{array} \right) \xrightarrow{R_3-R_1} \left(\begin{array}{ccc|c} 1 & 0 & -4 & -30 \\ 0 & 2 & 1 & 17 \\ 0 & 4 & 2 & 34 \end{array} \right) \xrightarrow{R_3-2R_2} \left(\begin{array}{ccc|c} 1 & 0 & -4 & -30 \\ 0 & 2 & 1 & 17 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

$$\xrightarrow{\frac{1}{2}R_2} \left(\begin{array}{ccc|c} 1 & 0 & -4 & -30 \\ 0 & 1 & 1/2 & 17/2 \\ 0 & 0 & 0 & 0 \end{array} \right) \xrightarrow{\text{RREF}} \begin{aligned} x &= -30 + 4z \\ y &= 17/2 - 1/2 z \\ z &= z \end{aligned}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -30 \\ 17/2 \\ 0 \end{pmatrix} + z \begin{pmatrix} 4 \\ -1/2 \\ 1 \end{pmatrix}$$

Name:

Key

5. a) Compute the inverse of each of the following matrices; or explain why an inverse does not exist.

$$A = \begin{pmatrix} 4 & 0 \\ 3 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

$$A^{-1} = \frac{1}{4} \begin{pmatrix} 1 & 0 \\ -3 & 4 \end{pmatrix} = \begin{pmatrix} 1/4 & 0 \\ -3/4 & 1 \end{pmatrix}$$

$$\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 4 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & 0 & 1 \end{array} \xrightarrow{R_2 - 4R_1} \begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -4 & 1 & 0 \\ 0 & 0 & -1 & 0 & 0 & 1 \end{array}$$

$$\xrightarrow{-R_3} \begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -4 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 \end{array} \Rightarrow B^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

b) Give an example of two matrices A and B such that A is invertible, B is not invertible, and $A + B$ is not invertible; or explain why this is not possible.

Many examples exist. For instance:

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix} \quad A + B = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad A + B = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$